Prateek Sinha

Chennai, India | ps826105@gmail.com | github.com/CzPhantom10 | +91 914211703

Education

SRM Institute of Science and Technology — Kattankulathur, Tamil Nadu
Bachelor of Technology in Computer Science and Engineering — *2023 – 2027*

- CGPA: 8.32
- Active participant in Hackathons and AI Research Projects

Experience

AI Researcher and Engineer — Jan 2024 – Present
SRM Institute of Science and Technology

- Fine-tuned GPT-2 to create **LawGPT** for Indian legal Q&A and judgment summarization
- Built a transformer-based language model from scratch using PyTorch
- Developed a Rust-based chatbot using Groq API for real-time LLM-based interaction
- Created a YOLOv8-based fire & smoke detection system for safety surveillance

Projects

LawGPT | *Python, PyTorch, GPT-2, HuggingFace* — May 2025

- Fine-tuned GPT-2 on Indian legal datasets (IPC, CrPC, judgments) for legal question answering
- Implemented LoRA and custom tokenizers for efficient training

LLM From Scratch | *Python, PyTorch* — Mar 2025

- Built a transformer architecture from scratch including self-attention, tokenization, and training loop
- Tested generation capability on small corpora and validated syntactic structure

Rust LLM Chatbot | *Rust, Groq API* — Feb 2025

- Developed a CLI-based chatbot in Rust using Groq's large language model API
- Optimized memory usage for low-latency response generation

Fire & Smoke Detection System | *YOLOv8, OpenCV, Python* — Dec 2024

- Designed a real-time video stream detection system for smoke and fire hazards
- Utilized bounding box tracking and confidence score filters for false-positive reduction

Technical Skills

Languages: Python, Rust, C, C++, SQL (MySQL)
AI/ML Frameworks: PyTorch, TensorFlow, LangChain, Transformers, OpenCV
Tools & Platforms: Git, Docker, Jupyter, VS Code
Domains: Generative AI, LLMs, Computer Vision, Real-time Systems